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**Thesis Title:** User Behaviour Mapping and AI-Powered Personalized Recommendations

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## Introduction

Traditional recommendation systems rely heavily on static historical profiles, failing to capture real-time consumer intent or solve cold-start limitations during active browsing sessions. While graph models like LightGCN capture structural signals, they struggle with severe data sparsity and a rigid accuracy-diversity trade-off. Current e-commerce literature separates precision-oriented models from diversity-focused frameworks. This thesis addresses this specific operational gap by developing the hybrid LGCNHS model. By merging physics-based resource allocation dynamics within a learned graph-neural network space, this adaptive framework utilizes granular implicit feedback signals to optimize personalization accuracy and list diversity simultaneously.

## Literature Review

Traditional collaborative filtering models (Konstan & Terveen, 2021) established user profiling based on historic, static data. However, modern e-commerce demands real-time adaptation. Recent advancements have shifted the focus toward session-based recommendation systems (SRSs) that map sequential item transitions within a single active session, capturing immediate, fleeting consumer intent rather than relying purely on long-term historical records.

While explicit ratings (like 5-star reviews) are notoriously sparse, implicit feedback provides a continuous stream of user behavior data. Studies by Wei, He, and Gan (2022) proved that integrating page dwell time as a high-density proxy signal significantly reduces noise and ambiguity in user clicks. Dwell time serves as a reliable indicator of genuine product interest, helping to filter out accidental clicks from actual engagement.

Graph Neural Networks, particularly LightGCN (He et al., 2020), revolutionized the field by modeling user-item interactions as high-order graph structures. However, these models still face the classic diversity-accuracy dilemma identified by Zhou et al. (2010). This laid the foundation for network-based physical resource allocation models, where algorithms like Probabilistic Spreading maximize accuracy, while Heat Spreading maximizes list diversity.

**Figure 2: The E-Commerce Accuracy-Diversity Dilemma Matrix**


## Methodology

**Framework & Data:** Structured using CRISP-DM on the MovieLens 20M dataset. Ratings are binarized ( \ge 3.5), and a dense subgraph is extracted via a 90th percentile quantile filter ( \epsilon_1=\epsilon_2=0.90 ). Latent features are built using a Truncated SVD (rank 62) concatenated with normalized structural node degrees.

**Model Architecture (LGCNHS)**

***Embeddings:*** fixed SVD matrices are mapped to a hidden space (d_h=64) using learnable linear projections ( W_u, W_o ).

***Propagation:*** K=3 layer LightGCN convolution aggregates multi-hop relational data using symmetric Laplacian normalization.

***Prediction:*** Layer embeddings are linearly combined (\rho_k = \frac{1}{k+1}) to generate a row-normalized allocation affinity matrix G = E_U E_O^T.

***Optimization:*** Minimizes pairwise Bayesian Personalized Ranking (BPR) Loss.

***Hybrid Spreading:*** Integrates physical Probabilistic (ProbS) and Heat (HeatS) resource propagation, modulated by the learned affinity values of G to break the accuracy-diversity trade-off.

**Training Extensions**

Enhanced with a Cosine Annealing Learning Rate Schedule (initialized at 1\times10^{-3}) and an Ensemble Scoring Strategy that blends global network co-interactions with local graph convolutional predictions

## Results

**Table 1: Comparison of Algorithm Performance at L = 30 (MovieLens)**

| Algorithm | P@30 | R@30 | F1@30 | NDCG@30 | H@30 | I@30 |
|---|---|---|---|---|---|---|
| ProbS | 0.467 | 0.205 | 0.231 | 0.519 | 0.477 | 0.392 |
| HeatS | 0.691 | 0.406 | 0.393 | 0.797 | 0.922 | 0.339 |
| HybridS | 0.631 | 0.371 | 0.350 | 0.730 | 0.908 | 0.391 |
| LightGCN | 0.582 | 0.317 | 0.317 | 0.667 | 0.795 | 0.418 |
| LGCNHS | 0.652 | 0.386 | **0.368** | 0.752 | 0.852 | 0.340 |

**Table 2: Comparison of Algorithm Performance at L = 50 (MovieLens)**

| Algorithm | P@50 | R@50 | F1@50 | NDCG@50 | H@50 | I@50 |
|---|---|---|---|---|---|---|
| ProbS | 0.430 | 0.296 | 0.294 | 0.495 | 0.446 | 0.359 |
| HeatS | 0.606 | 0.522 | 0.448 | 0.760 | 0.889 | 0.335 |
| HybridS | 0.540 | 0.464 | 0.389 | 0.683 | 0.878 | 0.376 |
| LightGCN | 0.517 | 0.432 | 0.377 | 0.637 | 0.750 | 0.376 |
| LGCNHS | 0.572 | 0.496 | 0.423 | 0.714 | 0.810 | 0.307 |

**Table 3: Comparison of Algorithm Performance at L = 100 (MovieLens)**

| Algorithm | P@100 | R@100 | F1@100 | NDCG@100 | H@100 | I@100 |
|---|---|---|---|---|---|---|
| ProbS | 0.354 | 0.506 | 0.350 | 0.507 | 0.386 | 0.314 |
| HeatS | 0.489 | 0.749 | 0.493 | 0.768 | 0.817 | 0.328 |
| HybridS | 0.414 | 0.631 | 0.408 | 0.659 | 0.807 | 0.332 |
| LightGCN | 0.421 | 0.649 | 0.423 | 0.647 | 0.682 | 0.318 |
| LGCNHS | 0.462 | 0.709 | 0.465 | 0.723 | 0.728 | 0.261 |

## Conclusion

**Empirical Findings**

The hybrid LGCNHS framework was successfully validated on the MovieLens benchmark, consistently outperforming traditional baselines across core accuracy metrics. While HeatS maintained maximum inter-user coverage ( ), the proposed model achieved the ultimate accuracy-diversity equilibrium. It recorded the lowest intra-similarity scores ( I = 0.261 at L=100 ), delivering highly diverse recommendations without sacrificing precision.

**Core Contributions**

The integration of a row-normalized allocation matrix ( G ) successfully bypassed the accuracy-diversity dilemma. The ablation study verified that removing the embedding layer (LGCNHS-e) significantly degraded precision, proving that BPR-trained linear projections are vital for high-fidelity affinity mapping. Additionally, custom training extensions, including a Cosine Annealing learning rate schedule and ensemble scoring, maximized convergence stability.

**Limitations & Future Outlook**

Aggressive 90th percentile filtering was implemented to manage computing constraints on CPU hardware. Future research will leverage multi-GPU setups to process larger subgraphs, integrate semantic e-commerce metadata via Pre-trained Language Models, and explore decentralized federated learning paradigms to support privacy-preserving multi-channel retail environments

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